

Kushal Koirala

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Professional Summary

Computational Chemist and PhD candidate in Bioinformatics and Computational Biology (UNC Chapel Hill) with 4+ years of experience in computer-aided drug design (CADD): structure-based drug design, molecular docking, molecular dynamics (MD) and enhanced-sampling simulations, free energy and binding kinetics, QSAR, ultra-large virtual screening, and machine learning (deep learning, graph/knowledge-graph models). Author of 8 peer-reviewed publications and 1 preprint, including 3 first or co-first-author works, with 2 additional first or co-first-author manuscripts in review or preparation and posters at ACS national meetings. Skilled in building reproducible HPC pipelines from target identification through hit identification and lead optimization.

Technical Skills

Computational Chemistry & Drug Design: Molecular docking (GNINA, Glide), molecular dynamics (AMBER, CHARMM, GROMACS), Gaussian accelerated MD (GaMD), free energy and binding kinetics, QSAR, pharmacophore modeling, high-throughput virtual screening (HTVS), fragment-based screening, structure-based and ligand-based drug design, hit-to-lead optimization

Cheminformatics & Structural Biology: RDKit, Schrödinger Suite (Maestro, Glide), AlphaFold, PDB data curation, MDAAnalysis, PyMOL, VMD, UCSF Chimera

Machine Learning & Data: PyTorch, TensorFlow, scikit-learn, transformers, diffusion models, graph neural networks and knowledge graphs (Neo4j, Cypher), Kedro, Seaborn, Pandas, NumPy

Programming & Infrastructure: Python, R, Bash, MATLAB, SQL; Linux/Unix, Git, Docker, High-Performance Computing (HPC), SLURM, Google Cloud, Jupyter

AI Tools: Anthropic Claude and OpenAI ChatGPT (prompt engineering)

Languages: English, Nepali, Hindi

Professional Experience

Graduate Research Assistant, Computational Drug Discovery

Aug 2023 – Present

Curriculum in Bioinformatics and Computational Biology, University of North Carolina at Chapel Hill

- Developed a knowledge-graph (KG) drug repurposing pipeline with a TensorFlow multilayer perceptron for drug–disease–target classification; nominated novel drug hypotheses for neurodegenerative diseases (first-author manuscript under review, ICLR 2027).
- Collaborated with the non-profit Every Cure on MATRIX, an AI platform for drug repurposing ([TIME Best Inventions 2025](#)); built and ran ML experiments with TensorFlow, scikit-learn, Seaborn, Kedro, and a biomedical KG to generate repurposing nominations across all known diseases. Co-author of the MeDIC resource supporting the platform.
- Engineered transformer and diffusion models (millions of parameters) trained on protein–ligand complexes for binding pose prediction and ligand affinity estimation.
- Built an automated pipeline for PDB data extraction, curation, and train/test splitting; curated ~190K protein–ligand structures into PURE, a benchmark for unbiased evaluation of docking and affinity models (co-first-author manuscript in preparation).
- Screened the Enamine REAL database (~14B compounds) against UppS in *Neisseria gonorrhoeae* using SMARTS-based pharmacophore filtering and HTVS; identified 356 early-stage hits for experimental validation and optimization.
- Built QSAR models for YCK2 kinase to rank candidate compounds for SAR optimization.
- Contributed to a machine-learning implicit solvation model for free energy calculations (arXiv, 2025).
- Competed in the CACHE (Critical Assessment of Computational Hit-finding Experiments) blind prediction challenges, nominating top 100 hits for multiple targets; placed 7th of 25 teams in CACHE Challenge #5.

- Developed PepBinding, a Python workflow combining peptide docking with peptide Gaussian accelerated MD (GaMD) to predict peptide–protein binding structures (co-first-author publication, *J. Phys. Chem. B*).
- Characterized binding mechanisms of small-molecule ligands selective for RNA single-G bulges with MD simulations; identified structural determinants of ligand selectivity (*Nucleic Acids Research*, 2025).
- Co-authored reviews on binding free energy and kinetics prediction (*J. Chem. Theory Comput.*; *Expert Opin. Drug Discov.*) and book chapters on accelerated MD for drug discovery.

Education

PhD, Bioinformatics and Computational Biology
University of North Carolina at Chapel Hill, Chapel Hill, NC

Aug 2023 – Dec 2027 (Expected)

Bachelor of Technology, Biotechnology
Kathmandu University, Kathmandu, Nepal

Jul 2014 – Jul 2018

Publications

Full list: [Google Scholar](#). *Co-first author.

Manuscripts Under Review & Preprints

1. **Koirala, K.**, DeLuca, M., Chung, C.-H., Sundar, S., Kapadia, S., Kaneva, V., Ding, E., Chirkova, R., Haendel, M. A., Bizon, C., Tropsha, A. (2026). Drug-Target-Disease Embeddings Improve the Accuracy of Drug Repurposing Model. *Under review, ICLR 2027*.
2. Kelestemur, E.*, **Koirala, K.***, Tropsha, A. PURE: Protein–ligand benchmark for Unbiased and Rigorous Evaluation. *Manuscript in preparation*.
3. Dey, R., Brocidiacano, M., **Koirala, K.**, Tropsha, A., Popov, K. I. (2025). Extending machine learning model for implicit solvation to free energy calculations. *arXiv preprint*. doi:10.48550/arXiv.2510.20103

Research Articles

1. DeLuca et al. (2025). Medicines, Diseases, Indications, and Contraindications (MeDIC): A Foundational Resource to Support Drug Repurposing. *Nucleic Acids Research*, gkaf1312. doi:10.1093/nar/gkaf1312
2. Ashizawa et al. (2025). Modeling Protein–Protein and Protein–Ligand Interactions by the ClusPro Team in CASP16. *Proteins: Structure, Function, and Bioinformatics*. doi:10.1002/prot.70066
3. Hegde et al. (2025). Mechanistic studies of small molecule ligands selective to RNA single G bulges. *Nucleic Acids Research*, 53(12). doi:10.1093/nar/gkaf559
4. Wang, J.*, **Koirala, K.***, Do, H. N., Miao, Y. (2024). PepBinding: a workflow for predicting peptide binding structures by combining peptide docking and peptide Gaussian accelerated molecular dynamics simulations. *J. Phys. Chem. B*. doi:10.1021/acs.jpcc.4c02047

Review Papers

1. Adediwura, V. A.*, **Koirala, K.***, Do, H. N., Wang, J., Miao, Y. (2024). Understanding the impact of binding free energy and kinetics calculations in modern drug discovery. *Expert Opinion on Drug Discovery*. doi:10.1080/17460441.2024.2349149
2. Wang, J., Do, H. N., **Koirala, K.**, Miao, Y. (2023). Predicting biomolecular binding kinetics: A review. *J. Chem. Theory Comput.* doi:10.1021/acs.jctc.2c01085

Book Chapters

1. Do, H. N., Wang, J., Joshi, K., **Koirala, K.**, Miao, Y. (2025). Gaussian Accelerated Molecular Dynamics in Drug Discovery. In *Computational Drug Discovery: Methods and Applications*, Wiley-VCH, 21–43. doi:10.1002/9783527840748.ch2

2. **Koirala, K.** et al. (2024). Accelerating molecular dynamics simulations for drug discovery. In *Computational Drug Discovery and Design*, 187–202, Springer US.

Conference Presentations

- **Koirala, K.**, et al. Accurate prediction of binding affinity from single structure. ACS Spring 2026 Meeting, Atlanta, GA, March 2026. (Poster)
- **Koirala, K.**, et al. Generation of novel drug repurposing hypotheses using LLM-augmented biomedical graph embeddings via multi-layer perceptron model. ACS Fall 2025 Meeting, Washington, DC, August 2025. (Poster)

Teaching Experience

Graduate Teaching Assistant, Statistical Modeling (BCB 720) Aug 2024 – Dec 2024
University of North Carolina at Chapel Hill

- Supported 10+ graduate students and led problem-solving sections.

Graduate Teaching Assistant, Biochemistry Aug 2021 – Jul 2023
University of Kansas

- Led laboratory sessions for 50 undergraduates, taught conceptual biochemistry, ran 3 problem-solving sections, and held office hours.
- Mentored 2 high school students on computational summer projects.

Experimental Research Experience

Molecular Lab Technologist Jun 2020 – Apr 2021
Decode Genomics and Research Center, Kathmandu, Nepal

- Frontline COVID-19 testing: sample preparation, DNA/RNA isolation, PCR, and gel electrophoresis.

Research Intern Apr 2019 – Dec 2020
Center for Health and Disease Studies (CHDS), Nepal

- Analyzed EGFR mutations in non-small cell lung carcinoma patients at Nepal Cancer Hospital and Research Center; set up laboratory space for microbial studies.

Undergraduate Researcher Mar 2018 – Sep 2018
Kathmandu University, Nepal

- Optimized production of high-fructose corn syrup for industrial application.

Leadership, Hackathons & Training

Steering Committee Member, BCB Department, UNC Chapel Hill Jun 2024 – Present

- Evaluated 10+ faculty candidates (CV review, research talks); contributed to 5 successful hires and provided the graduate student perspective on curriculum and program development.

Python Programming Tutor, How To Learn To Code (HTLTC) Jun 2024 – Present

- Mentored 10+ students from non-technical backgrounds in Python fundamentals; 100% course completion.

Hackathons and Training

- CMU x NVIDIA Federated Learning Hackathon for Biomedical Applications (2026).
- Bio-Agent Knowledge Graph Construction Challenge, NVIDIA x AWS (ongoing).
- High-Throughput Virtual Screening for Hit Finding and Evaluation, Schrödinger (2024).